MARKET MECHANISM

QUICK REVISION

1. Meaning of Equilibrium Price: Equilibrium Price is the price at which quantity demanded of a commodity is exactly equal to its quantity supplied. Another name for equilibrium price is market price.
2. Excess Demand : When the quantity demanded exceeds the quantity demanded there is an excess demand. The price this case.
3. Excess Supply: The supply o f a commodity is said to an excess supply when quantity supplied exceeds the quantity demand. The price will fall in this case.
4. Changes in Equilibrium Prices: Changes in equilibrium prices (or market prices) will occur as a result of changes in demand and supply.
5. Shift in Demand and Equilibrium Price:
(i) When supply remains constant, the upward shift of the demand curve will cause equilibrium price and quantity to increase. But the downward shift of the curve will lead to decrease in both equilibrium price and quantity.
(ii) When supply is perfectly elastic, the upward or downward shift of the demand curve will have no effect on the equilibrium price. It only causes equilibrium quantity to change.
(iii) When supply is perfectly inelastic, change in demand will affect the equilibrium price but not· the equilibrium quantity.
6. Shift in Supply and Equilibrium Price:
(i) When demand remains the same, equilibrium price falls and quantity increases with an increase in supply. But, if supply decreases, equilibrium price rises and quantity falls.
(ii) When the demand is perfectly elastic, change in supply will have no effect on the equilibrium price. But quantity will increase with increase in supply and decrease with decrease in supply.
(iii) When demand is perfectly inelastic, the rightward shift of the supply curve will reduce the equilibrium price. But a decrease in supply would increase the equilibrium price.
7. Simultaneous Change in both Demand and Supply:
(i) When increase in supply is equal to increase in demand, equilibrium price will remain unaffected.
(ii) When increase in supply is more than increase in demand, new equilibrium price will fall but equilibrium quantity will increase.
(iii) When increase in supply is less than increase in demand, new equilibrium price rises. The quantity bought and sold also increases.
(iv) When decrease in demand and decrease in supply are equal, equilibrium price remains the same but equilibrium quantity falls.
(v) When decrease in demand is more than decrease in supply, equilibrium price and equilibrium quantity both fall.
(vi) When decrease in demand is less than decrease in supply, equilibrium price rises but equilibrium quantity falls.
8. Effect of Change in both Demand and Supply in Reverse Direction:

(i) When increase in demand and decrease in supply are equal, equilibrium price rises but equilibrium quantity remains same.
(ii) When increase in demand is more than decrease in supply equilibrium price and equilibrium quantity both rises.
(iii) When increase in demand is than decrease in supply, equilibrium price rises but equilibrium quantity falls.
(iv) When increase in demand and increase in supply are equal, equilibrium price falls but equilibrium quantity remains same.
(v) When decrease in demand is more than increase in supply, equilibrium price and equilibrium quantity both fall.
(vi) When decrease in demand is less than increase in supply, equilibrium price falls but equilibrium quantity rises.

9. Special Cases of Market Equilibrium:

(i) When supply is perfectly elastic, then change in demand (increase or decrease in demand) does not affect the equilibrium price of the commodity. But equilibrium quantity rises in case of increase m demand and decreases in case of decrease in demand.
(ii) When supply is perfectly inelastic, any change in demand does not affect the equilibrium quantity. But equilibrium price rises in case of increase in demand and falls in case of decrease in demand.
(iii) When demand is perfectly elastic, then change in supply does not affect the equilibrium price. But equilibrium quantity rises in case of increase in supply and decrease in case of decrease in supply.
(iv) Quantity: When demand is perfectly inelastic, then change in supply does not have any effect on equilibrium quantity. But equilibrium price falls in case of increase in supply and rises of decrease in supply.

10. Applications of Tools of Demond and Supply:

Price Control Policies: Price control or regulation means fixation of price by law. At the controlled price, quantity demanded is not equal to the quantity supplied. The price may be fixed high or below the equilibrium price depending upon the situation.

(A) Maximum Price Policy or Price Ceiling: We know that if essential goods like sugar, rice, wheat, etc. are left to free market, poor people will not be able to buy them at the prevailing market determined price. Equilibrium price is presumed to be very high. In such a situation the Government fixes maximum affordable price for certain goods which is called price ceiling.

(B) Effects of Price Ceiling:

(i) If ceiling is above the equilibrium price then there will be the condition of excess supply and if it is fixed below the equilibrium price then there will be the condition of excess demand.
(ii) Allocation of Available Supply:

(a) First come first serve method,
(b) Distributing according to sellers' preferences,
(c) Rationing: It is a system of distribution of a specific quantity of a product at the price fixed by the Government. Under this system Government issues a ration card to each family 'which enables that family to buy a specified quantity of the product at a fixed price.

(iii) Black Marketing: Black marketing is a direct consequence of price controls.

(C) Minimum Support Price or Floor Price: The Government may fix the minimum price at which the sellers may sell a particular goods or service. Such price is known as floor price or minimum support price. Minimum support price fixed to assure the producers that they get a good price for their product.

Effects of Minimum Price:

(i) Surpluses (ii) Buffer stock (iii) Subsidies

1 MARK

QUESTIONS

1. The price at which both demand and supply are equal is called _____
2. If market price is less than the equilibrium price, the situation of arises. _____
3. If market price is more than the equilibrium price, the situation of arises. _____
4. If demand increases and supply remains same, equilibrium price _____
5. If supply increases and demand remains same, equilibrium price _____
6. In case of perfectly elastic demand, increase in supply will _____ the equilibrium price.
7. In case of perfectly inelastic supply, increase in demand will lead to _____
8. When both demand and supply increase by same percentage, equilibrium price _____
8. When demand increases more than increase in supply, equilibrium price _____
10. When demand decreases more than decrease in supply, equilibrium price _____
11. _____and _____ are major implications of price ceiling.
12. _____and _____ are important implications of minimum price.
13. If market price is less than the equilibrium price there will be competition amongst
14. If market price is more than the equilibrium price there will be competition amongst
15. When supply is _____ then change in demand will not affect the equilibrium price of the commodity.

ANSWERS

1. Equilibrium price	2. Excess demand	3. Excess supply	4 . Increases
5. Decreases	6. Not change	7. Increase in equilibrium price	8. Remains constant
9. Will increase	10 . Decreases	11. Rationing, Black marketing	12. Buffer stock, Subsidies
13. Buyers	14 . Sellers	15. Perfectly elastic	

B. TRUE OR FALSE

1. Excess supply leads to fall in equilibrium price.

Ans. [False] Excess supply leads to fall in market price but not in equilibrium price. During excess supply, market price is more than equilibrium price and in order to sell the excess stock, market price continues to fall till equilibrium price is achieved.

2. Change in supply will not change the equilibrium quantity in case of perfectly elastic demand.

Ans. [False] In case of perfectly elastic demand, equilibrium quantity will vary with change in supply but equilibrium price remains the same.

3. An increase in price of coffee will lead to rise in equilibrium price of tea.

Ans. [True] Increase in price of coffee will lead to increase in demand of tea which will lead to rise in equilibrium price.

4. If both demand and supply increase simultaneously, the equilibrium price will also change.

Ans. [False] If increase in demand is proportionately equal to increase in supply then equilibrium price will remain same.

5. Equilibrium quantity and equilibrium price remain same even with increase or decrease in demand in case of perfectly inelastic supply. .

Ans. [False] There will be no change in equilibrium quantity, but the equilibrium price will rise with increase in demand and will fall with decrease in demand.

6. If income of the consumer rises then equilibrium price of the inferior goods will also rise.

Ans. [False] An increase in income will decrease the demand for inferior goods, which will reduce the equilibrium price.

7. A simultaneous increase in demand and supply for a given commodity will result in more of the commodity being purchased.

Ans. [True] As both demand and supply increase, the commodity being purchased will rise.

8. With rise in input prices, equilibrium price of the commodity will increase.

Ans. [True] Equilibrium price will increase as supply will decrease due to rise in cost of production.

9. At a price higher than the equilibrium price, there is an excess demand.

Ans. [False] There is excess supply at price higher than equilibrium price.

10. If productivity of the commodity improves due to technological upgradation, then equilibrium price tends to increase.

Ans. [False] If productivity improves due to improvement in technology then the cost of production decreases, profit margin increases and supply will increase. In this case supply curve shifts to the right, demand remaining same equilibrium price decreases but equilibrium quantity increases.

11 . Excess supply of a commodity exists when its market price is greater than its equilibrium price.

Ans. [True] When market price is greater than the equilibrium price, then there is excess supply as market supply is more than the market demand. ·

12. Price ceiling is fixing the maximum price higher than the equilibrium price.

Ans. (False] As price ceiling is the maximum price fixed below the equilibrium price.

13. Floor price creates the problem of excess supply.

Ans. (True] Floor price is fixed higher than equilibrium price which creates the problem of excess supply.

C. MULTIPLE CHOICE QUESTIONS

MARKET MECHANISM

1. The price which determined with the equality of demand and supply is known as:

(a) equilibrium price (b) equilibrium quantity (c) balanced price (d) coordinated price

2. In a specific market and at a particular price, when market demand is greater than market supply it will be known as :

(a) excess supply (b) excess demand (c) excess price (d) excess quantity

3. When there is more supply than demand on a specific price in a given market, it will be known as:

(a) excess demand (b) excess supply (c) excess quantity (d) excess price

4 . Where market demand and market supply equals each other, what of the following is determined there?
(a) Equilibrium price (b) Equilibrium quantity (c) Both of these (d) None of these

5 . With the rise in price of factors of production, how supply of the product is affected?
(a) Supply curve shifts right
(b) Supply curve shifts left
(c) Demand curve shift right
(d) None of these

6 . With the favourable change in tastes of the product how equilibrium quantity is affected?
(a) Rightward shift in demand
(b) Leftward shift in demand
(c) Rightward shift in supply
(d) None of these

7 . What is the coordinating force between buyers and sellers?
(a) Cost of the product (b) Market price (c) Industry (d) Firm

8 . What will affect upon equilibrium price, when supply increases but demand remains constant?
(a) Rise in equilibrium price
(b) Fall in equilibrium price
(c) Equilibrium price remains constant
(d) None of these

9 . What affect will be there upon equilibrium price, when demand increases more than an increase in supply?
(a) Equilibrium price remains constant
(b) Equilibrium price rises
(c) Equilibrium price falls
(d) None of these

10. What will be affect of equilibrium quantity and equilibrium price when supply rises more than demand?
(a) Equilibrium price falls but quantity rises
(b) Equilibrium price and quantity remains constant
(c) Equilibrium price rises but equilibrium quantity falls
(d) None of these

11 . Whatever quantity sold on equilibrium price is known as:
(a) equilibrium quantity (b) balanced quantity (c) coordinated quantity (d) all of these

12. When demand increase but market supply remains constant, how equilibrium price is affected?
(a) Equilibrium price falls
(b) Equilibrium price rises
(c) No change in equilibrium price
(d) None of these

13. When demand for goods increases but supply decreases, how equilibrium price will be affected?
(a) Rises (b) Falls (c) Remains constant (d) None of these

14. Demand and supply increases simultaneously, how market price will be affected?
(a) Remains constant (b) Market price rises (c) Market price falls (d) Any of these

15. There is simultaneous increase in demand and supply but supply increases at proportionately greater rate. How equilibrium price is affected?
(a) Falls (b) Rises (c) Remains constant (d) None of these

16. When an increase in supply will lower the price, without affecting the level of demand:
(a) Then demand is perfectly elastic (b) Then demand is perfectly inelastic
(c) Then demand is less elastic (d) None of these

17. Without changing demand, when supply is decreased, how it will affect equilibrium price?
(a) Equilibrium price will rise (b) Equilibrium price will fall
(c) Equilibrium price remain constant (d) None of these

18. When price of a complementary goods falls, how it will affect the equilibrium quantity of related goods?
(a) Equilibrium quantity will remain constant (b) Equilibrium quantity will rise
(c) Equilibrium quantity will decrease (d) None of these

19. When price of tea rises what affect will be observed on demand for coffee?
(a) Demand curve for coffee will shift leftward (b) Demand curve for coffee will shift rightward
(c) No effect on demand curve for coffee (d) Any of these

20. There is an improvement in technique of production of a goods. How it will affect supply and equilibrium price?
(a) Supply will increase and equilibrium price falls (b) Supply will decrease and equilibrium price rises
(c) No effect on supply and equilibrium price (d) Any of these

21. There is rise in tax on production. How it affects on market supply and equilibrium price?
(a) Market supply falls and equilibrium price rises (b) Market supply rises and equilibrium price falls
(c) No change in market supply and equilibrium price (d) None of these

SIMPLE APPLICATION OF TOOLS OF DEMAND AND SUPPLY CURVES

22. Maximum controlled price refers to:
(a) floor price (b) minimum support price
(c) price ceiling (d) any of these

23. Which of the following is the effect of maximum ceiling price?
(a) Buffer stock (b) Food subsidies (c) Black marketing (d) None of these

24. Which of the following is a way of implementing maximum ceiling price?
(a) Bonus of farmers (b) Rationing system
(c) Procurement of food by Government (d) None of these

25. "Floor price" refers to:
(a) controlled price (b) maximum ceiling price
(c) minimum support price (d) any of these

ANSWERS

26. Main effect of floor price is :
(a) rationing system (b) buffer stock (c) black marketing (d) None of these

1. (a)	2. (b)	3. (b)	4. (c)	5. (b)	6. (a)	7. (b)	8. (b)	9. (b)	10. (a)
11. (a)	12. (b)	13.(a)	14.(d)	15.(a)	16. (b)	17. (a)	18. (b)	19. (b)	20. (a)
21. (a)	22. (c)	23.(c)	24.(b)	25. (c)	26. (b)				

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) Both (A) and (R) are true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : Floor price creates the problem of excess supply.
Reason (R) : Floor price is fixed higher than equilibrium price which creates the problem of excess supply as quantity supplied at this level is more than quantity demanded.

2. Assertion (A) : Equilibrium price is the price at which qua ntity demanded is equal to the quantity supplied.

Reason (R) : Equilibrium is the condition, once determined tends to persist in time.

3. Assertion (A) : Maximum support price creates the problem of excess demand.

Reason (R) : The negative implication of minimum support price is black marketing.

4. Assertion (A) : When demand is perfectly elastic then change in supply does not bring any change in equilibrium price.

Reason (R) : A perfectly elastic demand curve is a horizontal straight line parallel to 'X' axis.

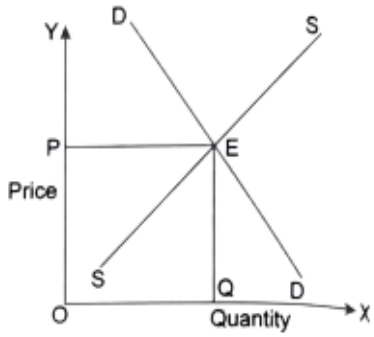
5. Assertion (A) : With increase in income, demand increases and the demand curve shifts to the right hand side.

Reason (R) : When demand increases keeping supply of the commodity same, the equilibrium price increases.

ANSWERS

1. (c)	2. (d)	3. (d)	4. (b)	5.(b)

2 MARK QUESTIONS

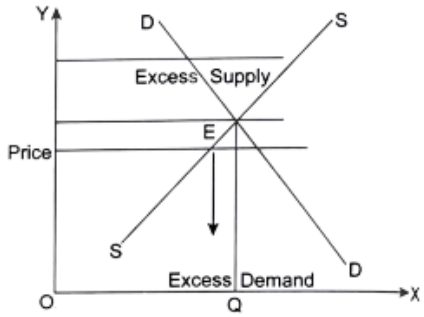


1 . Explain equilibrium price with the help of a diagram.

Ans. Equilibrium price is the price at which quantity demanded is equal to its quantity supplied. It can be shown with the help of adjoining graph. In the adjoining graph OP is the equilibrium price, E is the point of equilibrium and OQ is the equilibrium quantity where quantity demanded is equal to quantity supplied.

2. What will happen if price is less than equilibrium price?

Ans. If price is less than equilibrium price, it will be the condition of excess demand and the price will start increasing.



3. What do you mean by excess demand?

Ans. When market demand exceeds market supply of a commodity at a given price is known as excess demand.

4. Give the meaning of excess supply of a product.

Ans. Excess supply means market supply of the product is more than the market demand of the product at given price (graph given above)

5. How is equilibrium price affected by increase in demand?

Ans. When there is increase in demand keeping other things constant, equilibrium price rises.

6. How is equilibrium price affected by decrease in demand?

Ans. When there is decrease in demand keeping other things constant, equilibrium price decreases.

7. How is equilibrium price affected by decrease in supply?

Ans. When supply decreases keeping other things constant, equilibrium price increases.

8. How is equilibrium price get affected by increase in supply?

Ans. When supply of a commodity increases keeping other things constant, equilibrium price will decrease.

9. What do you mean by reserve price?

Ans. It is the price below which no producer wants to fix the price of his commodity.

10. What is the maximum price that a consumer will be prepared to pay for a commodity?

Ans. No consumer would like to pay price of the commodity more than its marginal utility. So he will be ready to pay a price for the commodity which is equal to its marginal utility.

11. What is the minimum price that a producer must get for his product?

Ans. The minimum price is governed by the cost of production of the commodity. No seller will be ready to accept a price for his product which is less than its marginal cost of production.

12. What is equilibrium point?

Ans. The point at which demand curve intersects supply curve is known as equilibrium point.

13. What is equilibrium quantity?

Ans. The quantity demanded and supplied at equilibrium price is known as equilibrium quantity.

14. What are the two variants of the policy of price control?

Ans. (i) Price ceiling (ii) Floor price.

15. What is black marketing?

Ans. Black marketing is a direct consequence of price controls. Black marketing implies a situation in which the controlled commodity is sold unlawfully under the table at a price higher than the lawfully enforced ceiling price. This situation arises largely because of the fact that: (i) number of potential consumers of the commodity is more than what can be served by the available supplies of the commodity and (ii) there are consumers who are willing to pay more than the ceiling price.

16. Why is price ceiling imposed?

Ans. Price celling is imposed in a situation when market equilibrium price is so high that it becomes out of reach for poor people and common consumers.

17. What is meant by minimum price or support price?

Ans. The minimum price is fixed by the Government in order to protect the interest of the producers.

18. When does change in demand have no impact on the price of the commodity?

Ans. When supply is perfectly elastic and supply curve is a horizontal straight line parallel to x-axis,

19. When does change in demand have no impact on equilibrium quantity?

Ans. When supply is perfectly inelastic and supply curve is a vertical straight line parallel to y-axis.

20. When does change in supply have no impact on equilibrium quantity?

Ans. When demand is perfectly inelastic and demand curve is a vertical straight line parallel to y-axis.

21. What is stable equilibrium?

Ans. Stable equilibrium is one which if displaced due to some small disturbances brings forces in operation which restores the initial equilibrium position.

3-4 MARKS QUESTIONS

1. What are the factors that determine the price equilibrium under perfect competition?

Ans. Determination of Equilibrium: Under perfect competition, the price of a commodity is determined by the general interaction of market forces of demand and supply in the industry.

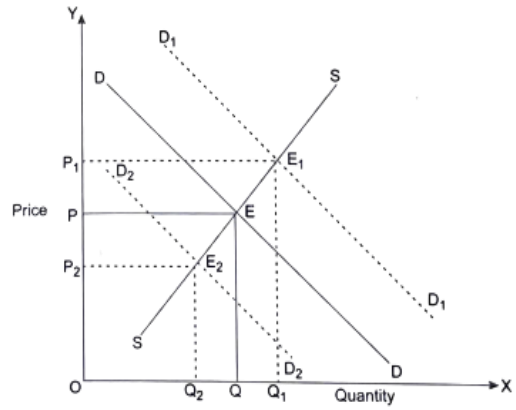
(i) Market Demand: It refers to the sum total of demand of commodity by all the buyers in the market.

(ii) Market Supply: It refers to the total supply of a commodity by all the firms in the market.

(iii) Equilibrium between Demand and Supply: In a perfectly competitive market, price is determined by interaction of market forces of demand and supply in the industry.

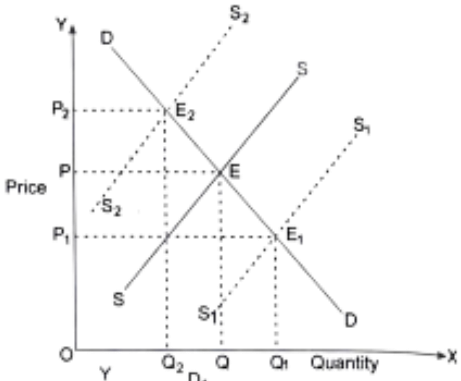
2. When supply remains unchanged, what is the effect of change in demand on price? Show it in the diagram.

Ans. When supply remains unchanged, the price increases with the increase in demand and price decreases with the decrease in demand. It is shown in the diagram alongside.



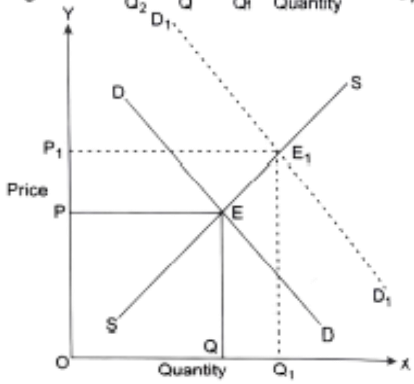
3. Demand remaining unchanged, what is the effect of change in supply on the price? Show it in the diagram.

Ans. When demand remains unchanged, the price decreases with the increase in supply and the price increases with the decrease in supply. It is shown in the diagram alongside.



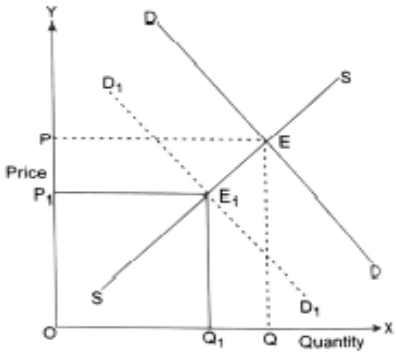
4. What happens to the equilibrium price when its demand increases (or demand curve shifts to the right)? Show it in the diagram.

Ans. If there is increase in demand (or rightward shift of the demand curve), equilibrium price would increase. In the diagram alongside, intersection between DD demand curve and SS supply curve determines OP as the equilibrium price. Now with the increase in demand, the demand curve shifts to the right, hence, equilibrium price increases from OP to OP1.



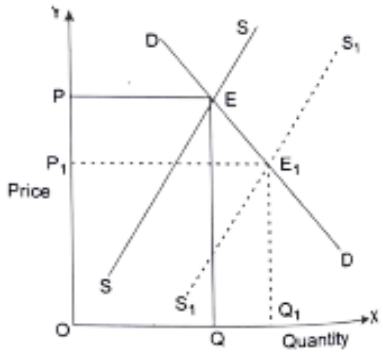
5. What happens to the equilibrium price when its demand decreases (or demand curve shifts to the left)? Show it in the diagram.

Ans. If there is decrease in demand (or leftward shift in the demand curve), the equilibrium price would decrease. In the diagram alongside, intersection between DD demand curve and SS supply curve determines OP as the equilibrium price. Now with the decrease in demand, the demand curve shifts to the left, hence, equilibrium price decreases from OP to OP1.

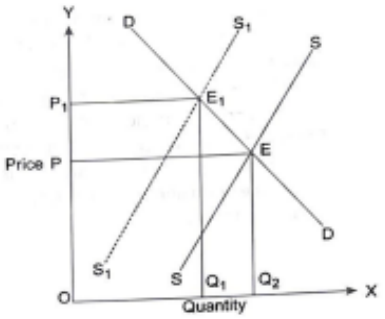


6. With the help of diagram, show the effect of an increase in supply (or rightward shift in the supply curve) on equilibrium price and quantity.

Ans. If there is increase in supply (or rightward shift in the supply curve), the equilibrium price decreases and quantity increases. In the diagram alongside, interaction between DD demand curve and SS supply curve determines OP as the equilibrium price and PE as the quantity. Now with the increase in supply, the supply curve shifts to the right, hence, equilibrium price falls from OP to op1 and quantity increases from PE to P1 E1.

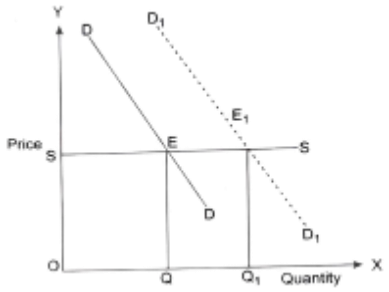


7. With the help of a diagram, show the effect of a decrease in supply (or leftward shift in the supply curve) on equilibrium price and quantity.



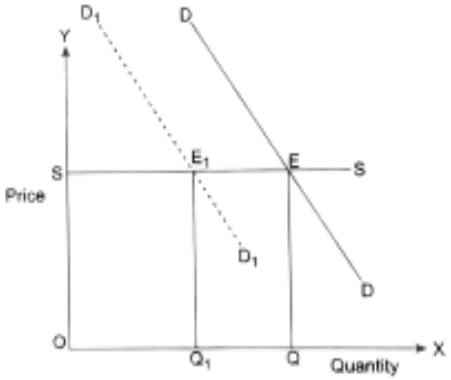
Ans. If there is decrease in supply (or leftward shift in the supply curve), the equilibrium price increases and quantity decreases. In the diagram alongside, interaction between DD demand curve and SS supply curve determines quantity. Now, with the decrease in supply, the supply curve shifts to the left, hence, equilibrium price rises from OP to OP1 and quantity decreases from PE to p1E1.

8. How will the equilibrium price and quantity of a commodity be affected with the increase in demand when its supply is perfectly elastic? Show in the diagram.



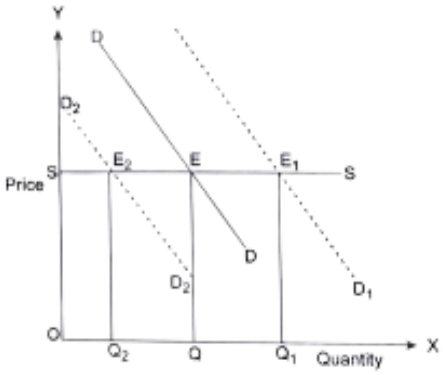
Ans. When supply is perfectly elastic, with the increase in demand, equilibrium price remains the same but equilibrium quantity increase. It is shown in the diagram alongside, DD demand curve intersects with the 55 supply curve at E which determines OS equilibrium price and OQ equilibrium quantity. Now, with the increase in demand new demand curve D1 D1 intersects with the SS supply curve at E1. Now equilibrium price remains the same as OS but quantity increases from OQ to OQ1.

9. How will the equilibrium price and quantity of a commodity be affected with decrease in demand when its supply is perfectly elastic? Show it in the diagram.



Ans. When supply is perfectly elastic, with the decrease in demand equilibrium price remains the same but equilibrium quantity decreases. In the diagram alongside, DD demand curve intersects with the SS supply curve at E which determines OS equilibrium price and OQ equilibrium quantity. With the decrease in demand new demand curve D1D1 intersects with the SS supply curve at E1. Now equilibrium price remains the same OS but quantity decreases from OQ to OQ1,

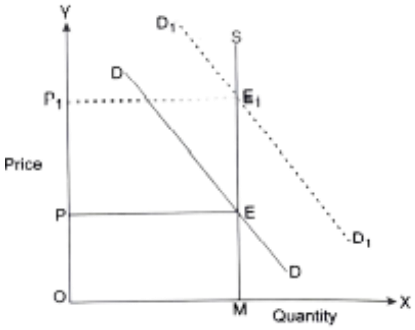
10. Show with the help of a diagram the effect of change in demand on equilibrium price when supply is perfectly elastic.



Ans. When supply is perfectly elastic, with the change in demand (or shifts in demand curve) price remains the same but quantity increases or decreases with the increase or decrease in demand.

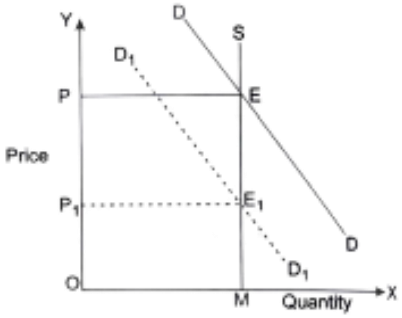
It is shown in the diagram alongside. With the increase or decrease in demand, price always remains the OS but quantity increases from OQ to OQ1 with the increase in demand and decreases from OQ to OQ2 with the decrease in demand.

11. When does the equilibrium quantity in a market remain unchanged with a change in demand? Show it with the help of a diagram.



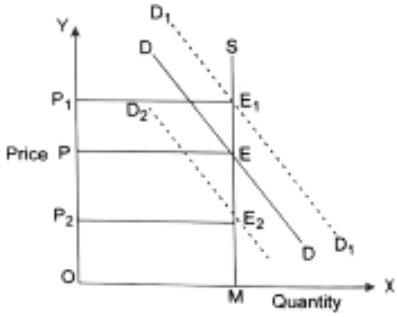
Ans. When supply is perfectly inelastic, with the increase in demand, equilibrium price increases but quantity remains the same. In the diagram alongside, DD demand curve and SM supply curve meet at E which determines OP equilibrium price and OM quantity. Now, with the Prid increase in demand, new demand curve D1 D_1 meets SM supply curve at E1, hence, equilibrium price increases from OP to OP, but quantity remains the same OM.

12. How will the equilibrium price and quantity of a commodity be affected with the decrease in demand (leftward shift in the demand curve) when its supply is perfectly inelastic? Show it in the diagram.



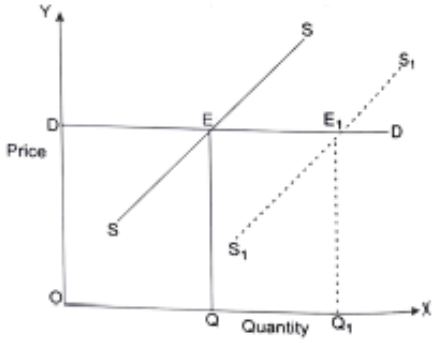
Ans. When supply is perfectly inelastic, with the decrease in demand, equilibrium price decreases but quantity remains the same. In the diagram alongside, DD demand curve and SM supply curve meet at E which determines OP equilibrium price and OM quantity. Now with the decrease in demand, new demand curve D1D1 meets SM supply curve at E_1, hence, equilibrium price decreases from OP to OP_1 but quantity remains the same OM.

13. How will the equilibrium price and quantity of a commodity be affected with the change in demand (or shifts in the demand curve) when its supply is perfectly inelastic? Show it in the diagram.



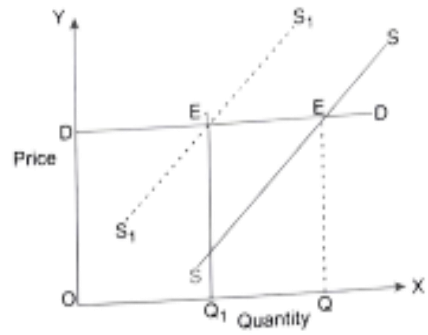
Ans. When supply is perfectly inelastic with the change in demand, (or shifts in demand curve), quantity remains the same but equilibrium price increases or decreases with the increase or decrease in demand. Diagram alongside shows that with the increase or decrease in demand, quantity always remains the same OM but equilibrium pries rises from OP to OP, with the increase in demand and falls from OP to OP2 with the decrease in demand.

14. How will the equilibrium price and quantity of a commodity be affected with the increase in supply (or rightward shift in the supply curve) when its demand is perfectly elastic? Show it in the diagram.



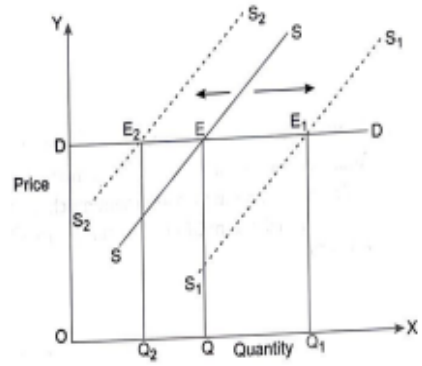
Ans. When demand is perfectly elastic, with the increase in supply, equilibrium price remains the same but quantity increases. In the diagram alongside, DD demand curve intersects the SS supply curve at E which determines OD equilibrium priced and OQ quantity. Now with the increase in supply new S1 S1 supply curve intersects DD demand curve at E_1, hence, equilibrium price remains the same OD but quantity increases from OQ to OQ1.

15. How will the equilibrium price and quantity of a commodity be affected with the decrease in supply (or leftward shift in the supply curve) when its demand is perfectly elastic? Show it in the diagram.



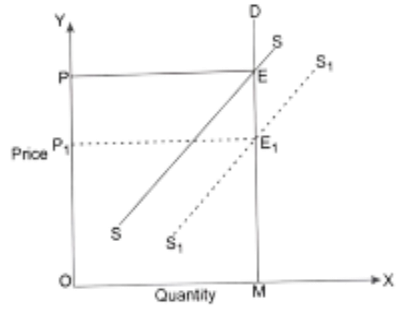
Ans. When demand is perfectly elastic, with the decrease in supply, equilibrium price remains the same but quantity decreases. In the diagram alongside, DD demand curve intersects the SS supply curve at E which determines OD equilibrium price and OQ quantity. Now, with the decrease in supply, new SS1 supply curve intersects DD demand curve at E1, hence, equilibrium price remains the same OD but quantity decreases from OQ to OQ1.

16. How will the equilibrium price and quantity of a commodity be affected with the change in supply (or shift in supply curve) when its demand is perfectly elastic? Show it in the diagram.



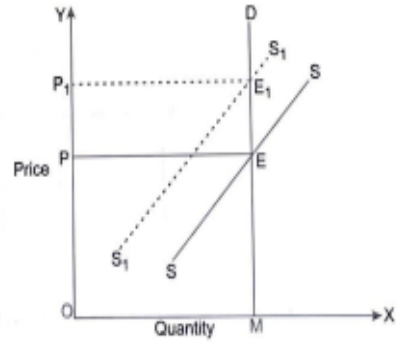
Ans. When the demand is perfectly elastic, with the change in supply (or shifts in supply curve), equilibrium price remains the same but quantity increases or decreases with the increase or decrease in supply. Diagram shows that with the increase or decrease in supply, equilibrium price always remains the same OD but quantity increases from OQ to OQ, with the increase in supply and decreases from OQ to OQ_2 with the in supply.

17. How will the equilibrium price and quantity of a commodity be affected with the increase in supply (or rightward shift in the supply curve) when its demand is perfectly inelastic? Show it in the diagram.



Ans. When demand is perfectly inelastic, with increase in supply, equilibrium price decreases but quantity remains the same. OP equilibrium price and OM quantity. Now with the increase in supply, new supply curve $S1S_1$ meets DM demand curve at E_1, hence, equilibrium price falls from OP to OP, but quantity remains the same OM.

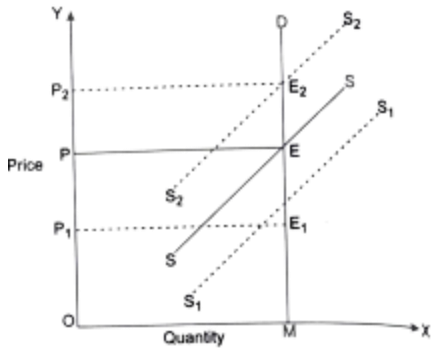
18. How will the equilibrium price and quantity of a commodity be affected with the decrease in supply (or leftward shift in the supply curve) when its demand is perfectly inelastic? Show it in the diagram.



Ans. When demand is perfectly inelastic, with the decrease in supply, equilibrium price increases but quantity remains the same. In the diagram alongside, DM demand curve and SS supply curve meet at E which determines OP equilibrium price and OM quantity. Now with the decrease in supply, new supply curve S_2S_1 meets DM demand curve at E_1, hence, equilibrium price rises from OP to OP, but quantity remains the same OM.

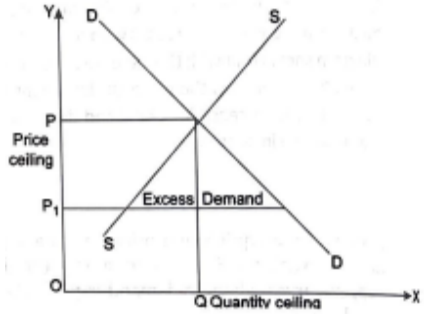
19. How will the equilibrium price and quantity of a commodity be affected with the change in supply (or shift in supply curve) when its demand is perfectly inelastic? Show it in the diagram.

Ans. When demand is perfectly inelastic, with the changes in supply, quantity remains the same but price decreases or increases with the increase or decrease in supply. Diagram alongside shows that with the increase or decrease in supply, quantity remains the same OM but equilibrium price decreases from OP to OP, with the increase in supply and price increases from OP to OP2 with the decrease in supply.



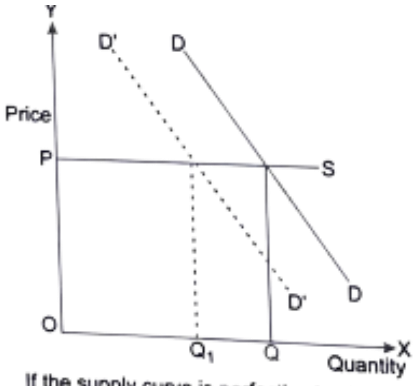
20. Explain the meaning of price ceiling with the help of a diagram.

Ans. The price ceiling is the maximum legal price of a commodity which is fixed by the Government much below the equilibrium price. In the given diagram OP, is the ceiling price which is below equilibrium price OP.

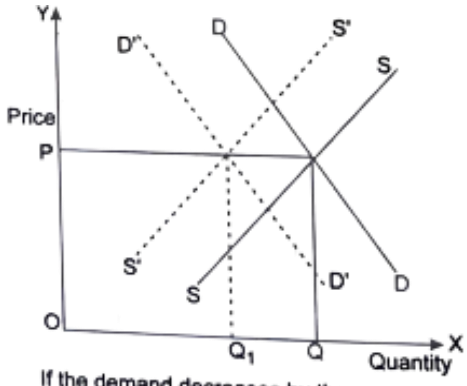


6-8 MARKS QUESTIONS

1. Show that a decrease in demand does not always lead to a fall in the price of a commodity. Ans. Under the following conditions decrease in demand may not lead to a fall in the price of a commodity:

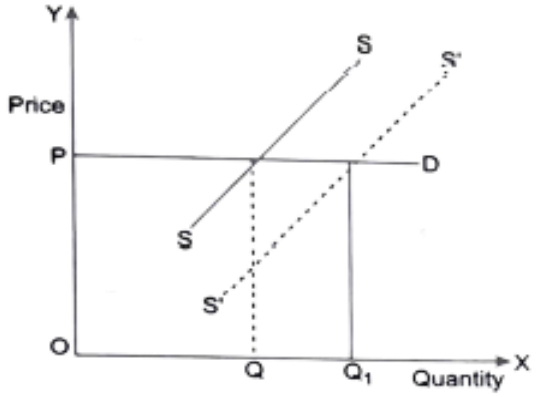


If the supply curve is perfectly elastic demand

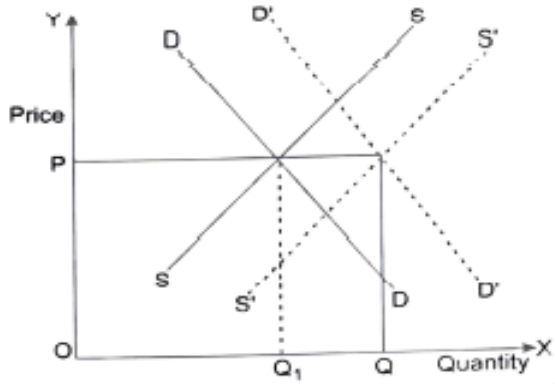


If the demand decreases by the same proportion as the decrease in supply

2. Show that an increase in supply does not always lead to a fall in the price of the commodity. (Use diagrams.)



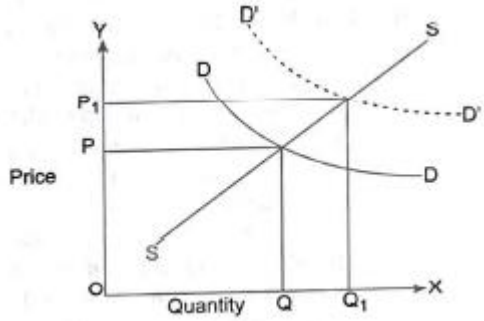
If the supply curve is perfectly elastic demand



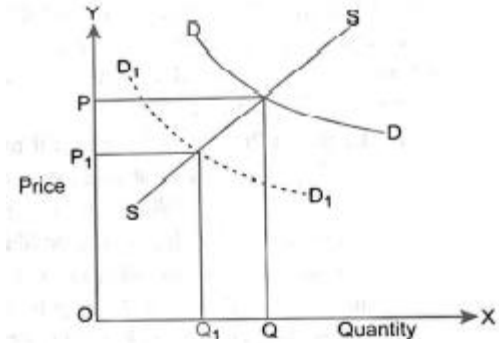
If the demand increases by the same proportion as the increase in supply

3. How does an increase in the income affect the equilibrium price of a product if it is a: (i) normal commodity and (ii) inferior commodity?

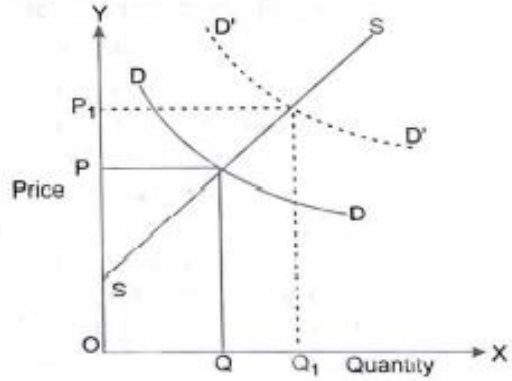
Ans. (i) With increase in income the demand curve would shift to the right, for normal goods the equilibrium price would increase from OP to OP1.



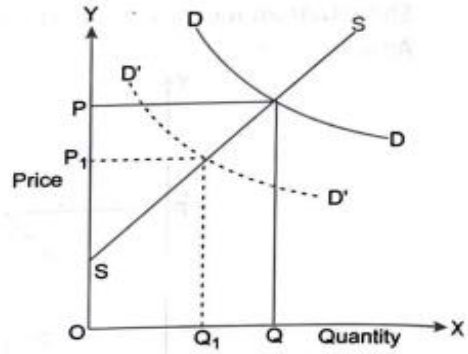
(ii) With increase in income, if it is an inferior goods, the demand curve would shift to the left and equilibrium price would fall from OP to OP1.



4. Trace the effect of demand shift on equilibrium price and quantity. Ans. When Demand Increase: When demand increase the equilibrium price increases from OP to OP, and equilibrium quantity increases from OQ to OQ1.

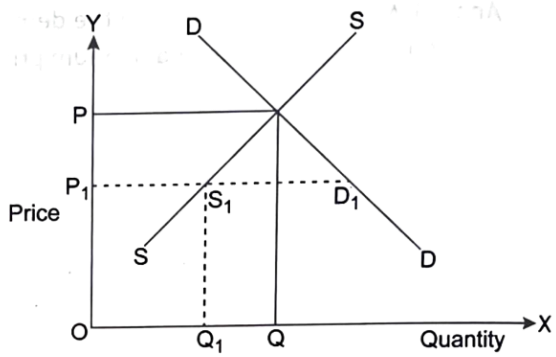


Ans. When Demand Decreases: When demand, decreases, the equilibrium price falls from OP to op1 and equilibrium quantity falls from OQ to OQ1.



5. Show with the help of a diagram how rationing and black marketing can emerge in a price control system.

Ans. Sometimes the Government feels that if the prices of necessary items like wheat, sugar, rice are let to the play of free market entirely, the price would rise so high that the poor would not be able to afford to buy these goods. Therefore, in such cases the Government fixes the price below the equilibrium price, say at OP_1. OP_1 is called the control price. At the control price the quantity demanded is P_1D_1 and the quantity supplied is P_1S_1. Therefore, there would be some people whose demand would not be satisfied. This implies:



(i) There has to be some rationing. Rationing means putting an upper limit on the amount that can be purchased within a given period. This way, though the entire demand of all the people will not be met but at least part of the demand of all the individuals would be met.

(ii) Since, there is a shortage of the commodity at the control price, there will always be some buyers who would be willing to pay a price Price higher than one fixed by Government to obtain the desired quantity of goods. This gives rise to a black marketing. (A black marketing is defined as a situation in which goods are sold at more than the price fixed by the Government by law.)

6. What is price ceiling? Explain the effect of price ceiling.

Ans. Maximum Price Policy or Price Ceiling: We know that if essential goods like sugar, rice, wheat, etc. are left to free market, poor people will not be able to buy them at the prevailing market determined price. Equilibrium price is presumed to be very high. In such a situation the Government fixes maximum affordable price for certain goods which is called price ceiling.

Effects of Price Ceiling:

1. Effects on Price and Quantity: If price ceiling is done above the equilibrium price then it has no effect on price and quantity. At a higher price there will be a condition of excess supply which will reduce the price and quantity of the commodity and it will rest at the equilibrium price which is OP. If the price is fixed below the equilibrium price then there will be a situation of excess demand which means there is a shortage of the commodity in the market.

Allocation of Available Supply: If at controlled price, supply is less than demand then every consumers' need can not be satisfied, and price also can not be increased. In that case there arises the problem of distributing the limited supply among all the consumers. Some of the methods of distributing supply are:

(i) First come first serve method.
(ii) Distributing according to sellers' preferences.
(iii) Rationing: It is a system of distribution of a specific quantity of a product at the price fixed by the Government. Under this system Government issues a ration card to each family which enables that family to buy a specified quantity of the product at a fixed price.
2. Black Marketing: Black marketing is a direct consequence of price controls. Black marketing implies a situation in which the controlled commodity is sold unlawfully under the table at a price higher than the lawfully enforced ceiling price. This situation arises largely because of the fact that: (i) number of potential consumers of the commodity is more than what can be served by the available supplies of the commodity and (ii) there are price controlled by the authorities will invariably arise since there are consumers who are willing to pay more than the controlled price.

7. What do you mean by floor price? Explain the effects of it.
Ans. Minimum Support Price or Floor Price: The Government may fix the minimum price at which the sellers may sell a particular goods or service. Such price is known as floor price or minimum support price. Minimum support price is fixed to assure the producers that they get a good price for their product. For example, in India the Government fixes up the minimum support price for several agricultural products like wheat, rice, jawar, bajro, etc.
Effects of Minimum Price: There will be following effects of floor price:
(i) Surpluses: The quantity actually bought and supplied will shrink because of less demand as a direct consequence of price support; as a result, large chunk of producers' stock will remain unutilized which is known as surplus.
(ii)Buffer Stock: The Government purchases the surplus product from farmers and maintains a buffer stock. Buffer stock is used to avoid the fluctuations in prices by releasing the stock at the time of excess demand and higher prices.
(iii) Subsidies: The Government purchases the product at the support price and sells the product to the consumers below its cost of procurement. The difference between the cost price and market price is borne by the Government.

QUESTIONS BASED ON DIFFERENTIATION

1. Differentiate between excess demand and excess supply.
Ans. The situations of excess demand and excess supply: A comparison is as:

S.No	Basis	Excess Demand	Excess Supply
1	Disequilibrium between demand and supply	Quantity demanded of a commodity exceeds its quantity supplied (i.e., Qd > Qs).	Quantity supplied (Os) exceeds demanded (Qd) (i.e., Q5 > Od)
2	Causes	An increase in demand due to rise in people's income or increase in the price of a substitute goods. Given the demand for a commodity, the excess demand may also arise when there is decrease in supply.	An increase in supply due to technological improvements or fall in wages or price of other inputs. Given the supply of a commodity, excess supply may also be created as a result of decrease in its demand.
3	Effect	It causes price to rise.	It causes price to fall.

CASE BASED QUESTIONS

CASE 1

Equilibrium is a state of rest or balance. Market equilibrium is determined with the help of demand and supply. Demand and supply forces are known as invisible hands of the market. These forces of demand and supply keeps on fluctuating according to the market conditions. These fluctuations in the market create the conditions of excess supply and excess demand. But these conditions of excess demand and supply are not permanent they get corrected and after some time reach to the condition of equilibrium again.

Answer the following questions based on the above paragraph :

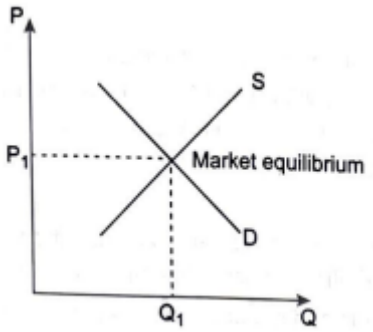
1. Define excess supply.
2. Explain any two implications of excess supply.
3. What do you mean by stable equilibrium?
4. Draw the graph to show the market equilibrium.

Ans. 1. A situation in which the market supply of a commodity is greater than the market demand for it, thus causing its market price to fall. .

2. Excess supply will cause price to fall, and as price falls producers are willing to supply less of the good, thereby decreasing output.

3. Stable equilibrium is one which if displaced due to some small disturbances brings forces in operation which restores the initial equilibrium position.

4.



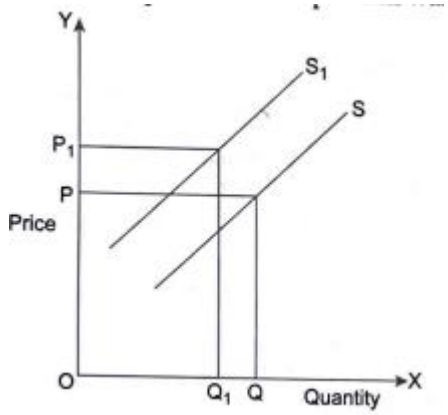
CASE 2

Due to too much rainfall in some states of northern India the wheat crop got damaged in 1998. Since wheat ls a staple food of northern India the demand for it remained the same but due to flood situation production of wheat was very low and price started increasing. Then government of India intervenes and provides help to the consumers of wheat.

Answer the following questions based on the above paragraph :

1. Draw the graph to show the above situation.
2. What step you think that government had taken to help the consumers of wheat?
3. What will be the impact of that step which was taken by the government in the above situation?

Ans. 1.



2. In such a situation the government fixes maximum affordable price for certain goods which is called price ceiling.

3. Effects of price ceiling:

(i) Effects on price and quantity: If price ceiling is done above the equilibrium price then it has no effect on quantity. At a higher price there will be a condition of excess supply which will reduce the price and quantity of the price and commodity and it will rest at the equilibrium price. If the price is fixed below the equilibrium price then there will be a situation of excess demand which means there is a shortage of the commodity in the market.

(ii) Black marketing: Black marketing is a direct consequence of price controls. Black marketing implies a situation in which the controlled commodity is sold unlawfully under the table at a price higher than the lawfully enforced celling price. This situation arises largely because of the fact that (a) no. of potential consumers of the commodity is more than what can be served by the available supplies of the commodity and (b) there are consumers who are willing to pay more than the ceiling price. Thus black marketing of a commodity whose price has been controlled by the authorities will invariably arise since there are consumers who are willing to pay more than the controlled price.

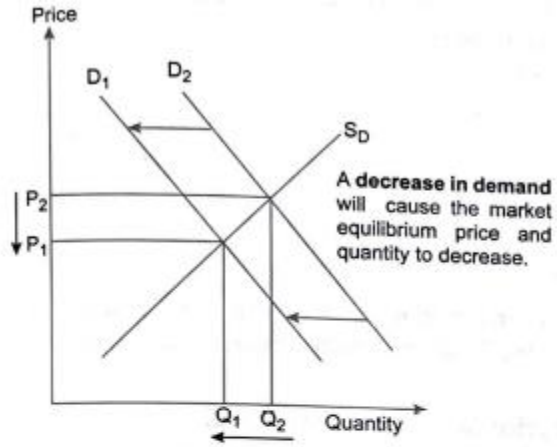
CASE 3

Due to COVID-19 pandemic the aggregate demand of the people in the entire world has gone down. People prefer spending money only on necessities and demand for luxuries and comforts got postponed. Since demand for luxuries and comforts is not much then the producers producing these things started producing few necessities also as their demand will always be there.

Answer the following questions based on the above paragraph

1. Draw the graph to show the decrease in demand and its impact on equilibrium price and quantity.
2. Show the impact of increase in the production of necessities with the help of graph.

Ans. 1. Market Equilibrium:



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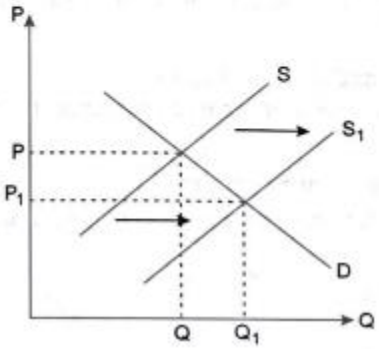


QUESTIONS WITH HIGH DIFFICULTY LEVEL

1. What will be the effect on equilibrium price and equilibrium quantity, when: (i) Income increases in case of normal goods; (ii) Price of complementary goods increases?

Ans. (i) Income Increases in Case of Normal Goods: If the income of consumers increases then they will be able to spend more money on normal goods, assuming no change in other factors. As a result, demand for normal goods will increases and the demand curve will shift towards right. Since, an increase in the income does-not have any impact on supply, the supply curve remains unchanged. Both equilibrium price and equilibrium quantity rises.

(ii) Price of Complementary Goods Increases: When price of complementary goods increases, keeping other factors constant, then demand for the given commodity decreases since it becomes relatively expensive to consume the two commodities (the given commodity and· its complement) together. It shifts the demand curve towards left. Since, an increase in the price of complementary goods _does not have any impact on supply, the supply curve remains unchanged. Both equilibrium price and equilibrium quantity falls.

2. What will be the effect on equilibrium price and equilibrium quantity, when: (i) Number of firms increases; (ii) Price of inputs increases?

Ans. (i) Number of Firms Increases: When number of firms increases, keeping other factors constant, then total supply in the market will also increase due to more producers Producing the commodity. It shifts the supply curve towards right. Since, an increase in number of firms does not have any impact on demand, the demand curve remains unchanged. Equilibrium price falls but equilibrium quantity rise.

(ii) Price of Inputs Increases: When price of inputs increase, assuming change in other factors, then the cost of production rises. As a result, supply decreases due to fall in _the profitab1hty level. It shifts the supply

curve towards left. Since, an increase in the price of inputs does_ not have any impact on demand, the demand curve remains unchanged. Equilibrium price rises but equilibrium quantity falls.

3. "Demand and supply are like two blades of a pair of scissors". Comment.

Ans. The given statement is correct. Both the blades of a pair of scissors are equally important to cut a pieceof cloth. Similarly, both demand and supply are needed for determining price in the market. There is no use for demand for a product if there is no supply for the product and supply is not needed if there is no demand for the product. One of the two may play more active role in price determination in the short-run. But both are needed to determine the price in the long-run.

4. Mention the various cases in which equilibrium price remains same.

Ans. The equilibrium price remains same in the following cases:

(i) When increase in demand is equal to increase in supply.
(ii) When decrease in demand is equal to decrease in supply.
(iii) When demand increases and supply is perfectly elastic.
(iv) When demand decreases and supply is perfectly elastic.
(v) When supply increases and demand is perfectly elastic.
(vi) When supply decreases and demand is perfectly elastic.

5. How does an increase in the price of a substitute goods in consumption affect the equilibrium price?

Ans. If the price of a substitute goods increases then the demand for the commodity increases and demand curve shifts to the right and equilibrium price increases.

6. How does a favourable change in taste affect the market price and quantity exchanged?

Ans. A favourable change in taste would shift the demand curve to the right and the equilibrium price and quantity increase.

7. How does a cost saving technological progress affect the market price and the quantity exchanged?

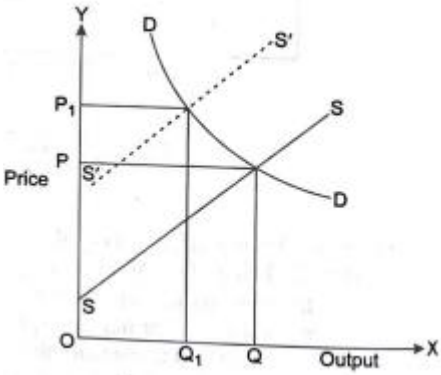
Ans. A cost saving technological progress would lead to a rightward shift of the supply curve and hence the market price would fall and quantity would increase.

8. How does an increase in excise tax rate affect the market price and the quantity exchanged?

Ans. An increase in excise tax rate leads to a leftward shift of the supply curve and hence it leads to a higher price and less quantity.

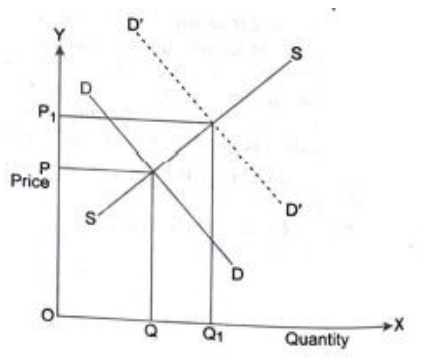
9. A severe drought result in a drastic fall in the output of wheat. Analyse how will it affect the market price of wheat?

Ans. A fall in the output of wheat would lead to a leftward shift of the supply curve. So the market price of wheat would rise from OP to OP1.



10. New discoveries of oil reduce the price of petrol and diesel. Explain their effects on the market for new cars.

Ans. Since, cars and petrol are complementary goods, with a reduction in the price of petrol, the demand for cars would increase and the demand curve for cars would shift to the right. As a result, the equilibrium price increases from OP to OP, and quantity demanded would increase from OQ to OQ. So the market for new cars Price would increase.



WORKSHEET FOR EVALUATION

1. Using diagram, discuss the effect on equilibrium price and quantity in the following cases:
(i) Increase in demand is equal to the decrease in supply.
(ii) Increase in demand is more than decrease in supply.
(iii) Increase in demand is less than decrease in supply.
(iv) Increase in supply is equal to decrease in demand.
(v) Increase in supply is more than decrease in demand.
(vi) Increase in supply is less than decrease in demand.

2. If there is excess supply at a giver price, then how will the equilibrium price be reached? Explain it with diagram. **(2 Marks)**

3. The market demand and market supply of a commodity is given below: **(4 Marks)**

Price	Demand	Supply
1	30	10
2	25	15
3	20	20
4	15	25
5	10	30

On the basis of the above table, answer the following questions:
(i) What will be the equilibrium price of the commodity?
(ii) What will be the equilibrium quantity demanded and supplied at the equilibrium price?
(iii) What will happen, if price is more than the equilibrium price?
(iv) What will happen, if price is less than the equilibrium price?

4. At a price of 5 per chocolate, the demand is 70 chocolates and the supply is 50 chocolates. What is likely to be its effect on the price of chocolates? **(2 Marks)**

5. What will be the effect on equilibrium price and equilibrium quantity of telephone instruments, if China exports a large number of telephone instruments to India. **(2 Marks)**

6. Explain the process of fixation of minimum and maximum price. **(3 Marks)**

7. Show the implication of fixation of floor price. **(3 Marks)**

8. What is rationing? **(2 Marks)**

9. Suppose the demand for jeans increases. At the same time, because of an increase in price of cotton, the supply of jeans decreases. How will it affect the price and quantity sold of jeans? **(3 Marks)**

10. In the budget of 2015-16, the indirect tax on coffee is increased. If all other things remain unchanged, how will it affect the market price of coffee? **(3 Marks)**

11. Show the effect of following on equilibrium price and equilibrium quantity: **(3 Marks)**

(i) Increase in demand is more than decrease in supply.

(ii) Increase in supply is less than decrease in demand.

12. Suppose the price control on wheat is lifted. How will it affect the price and quantity consumed of wheat? **(2 Marks)**

13. What is the main reason for the fixation of minimum support price and maximum support price? **(2 Marks)**

14. Who gets the benefits of price ceiling and floor price? **(2 Marks)**

15. For the fixation of price what is more important: demand or supply? **(4 Marks)**